Do Now · retrieval

Advanced Big O and space complexity

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does O(log n) indicate?

Q2

What is auxiliary space?

Q3

Is Big O an exact millisecond prediction?

Q4

A sorted array of length64 is reduced by half after each unsuccessful search step. How many halvings reach length1?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify the dominant operation — Single full loop grows O(n); two full nested n loops grow $O(n^2)$. Repeated halving grows O(log n).

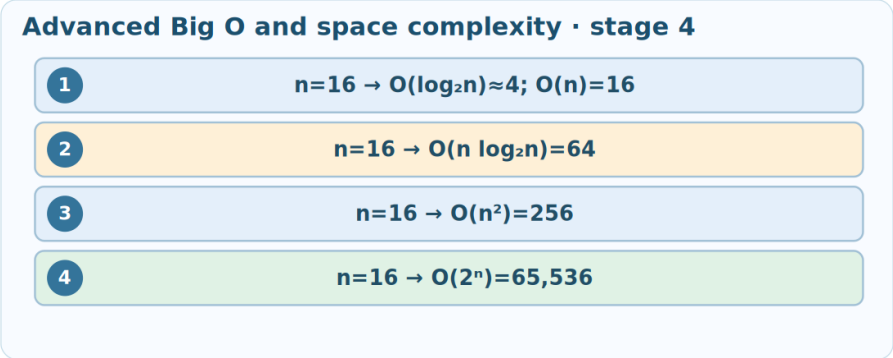
Core idea 2

Separate time and space — Counting two scalar variables uses O(1) auxiliary memory even if repeated n^2 times; copying n elements uses O(n) extra storage.

Core idea 3

Interpret asymptotics — Big O describes a bound on growth, not an exact runtime. Constants, cache effects and data size still matter.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

asymptotic — Behaviour as input size grows.

Big O — Asymptotic upper bound.

auxiliary space — Storage additional to input.

polynomial — Growth such as n^2.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

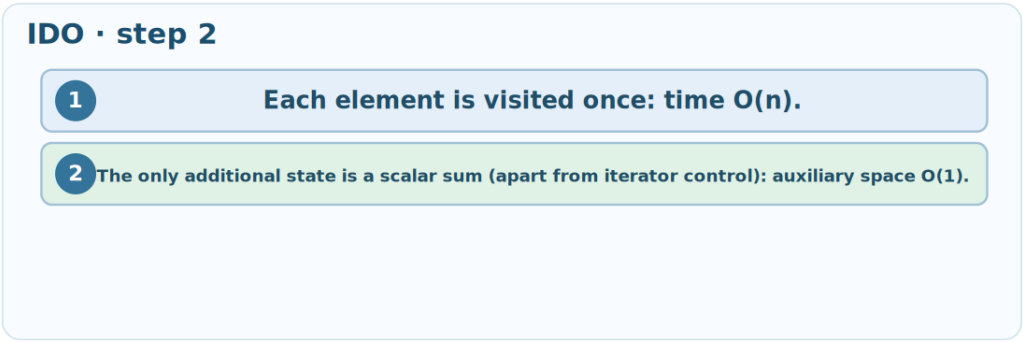
Find worst-case time and auxiliary space for summing n array elements into one accumulator.

Follow the worked method

1. Predict first — Each element is visited once: time O(n).

2. Explain the method — The only additional state is a scalar sum (apart from iterator control): auxiliary space O(1).

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Copy performs a constant amount of work n times, so time grows O(n).

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Classify loops running once, n times, n^2 times and repeatedly halving n. [4]

Question 2

A function makes a new copy of every one of n elements. Identify time and auxiliary space. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare bubble sort and merge sort at large n using time and space trade-offs. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An $O(n^2)$ algorithm must always use $O(n^2)$ memory.

Use key vocabulary from memory

1. Explain asymptotic in your own words.

2. Explain Big O in your own words.

3. Explain auxiliary space in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.